

Theoretical and Experimental Study of the Dynamic Behaviour of a Double-Cracked Beam

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Abstract. In recent years a significant amount of effort has been devoted to the development of nondestructive testing techniques for damage detection in structures. It is well-known that the dynamic response of a structure changes when it is subject to damage. Given a relationship between the dynamic property changes and the damage, measurement of the changes can be used as a diagnostic tool. When the damage of a structure is due to the presence of cracks, the measurement of the changes can be used to detect and quantify the extent of the damage. Previous research has revealed that both the location and quantification of damage is possible when a single crack is present. If the structure is cracked in at least two places, the identification problem becomes decidedly more complex. In fact, for two cracks, the solution depends on four variables, the position and the size of each crack. The aim of this paper is to validate two analytical models of a cantilever beam with two cracks using experimental data obtained from a real structure and to perform a systematic study of the analysis and experimentation by correlating the crack location, the crack size and the corresponding changes in modal parameters.

1 Introduction

The occurrence of damage in a structure causes changes in the dynamic properties of the structure. Given a relationship between the dynamic property changes and the damage, measurement of the changes can also be used as a diagnostic. Vibration diagnosis, as a non-destructive detection technique, has become of greater importance in recent years. Many studies have been devoted to the development of nondestructive testing techniques for detection of crack-induced damage in a structure. Structures may develop cracks as a result of some strong excitations, such as earthquakes, blast loading, wind load etc. Diagnosis of fracture damage is thus essential in assessing the reliability of the structure. Vibration investigation of a damaged structure is one approach for fault diagnosis. Several authors have investigated the natural frequency

variation due to propagating cracks, carrying out intensive experimental programmes. One crack models (with slots or cracks obtained with the aid of the three points bending technique) were studied for clamped-free, free-free beams, [1], [2], [3], [4], [5], [6]. For further studies and validation of damage detection methods, one needs experimental data for multiple cracks cases. In this paper, we present the experimental results obtained for beams with two slots and compare the results with those obtained using a simple finite element model and a sensitivity based modal model developed for two cracks, in order to evaluate the merits of the methods for determining the incursion of the damage in the structure.

The paper is organised as follows: the second section describes briefly the theoretical models, the third section presents the experimental procedure and set up; a comparison between analytical and experimental data is made in the fourth section. Finally the fifth section concludes the paper.

2 Theoretical Models

The finite element model presented in [7], [8] and a sensitivity based modal model for a beam with two cracks [9], [10], [11], [12], are used for comparison with the experimental data. For the finite element model, it is considered that a transverse crack in a structural member introduces a local flexibility due to the strain concentration in the vicinity of the crack tip under the load and complete flexibility matrices were derived and verified by experiments for different case studies. The modal model relates the magnitude of the crack with an equivalent linear spring term using fracture mechanics.

3 Test Beams and Experimental Procedure

The dynamic behaviour of a clamped-free beam was analysed. The experimental procedure was carefully programmed in order to guarantee a high level of accuracy and was carried out for several different combinations of cracks and depths. The cracks were simulated using electron discharge machining with a wire diameter of 0.10 mm and the measurements were performed for in plane flexural vibrations. The width of the crack was about 0.13 mm and experimental tests have assured that the opening-closing process typical for a true crack does not occur in this case. The dimensions of the beam, cross-section and length were chosen in order to allow for accurately determining of the first four natural frequencies in the range up to 1000 Hz and for accurate positioning of the transducers. The test specimens were made of steel (C70) with 0.8 m length and a cross section 0.020 m x 0.020 m. In order to avoid the influence of small geometric differences in the test beams, the measurements were performed first on the nonslotted beam and then the results were compared with that ones obtained after the machining the slots. A total of 15 different damage scenarios (listed in Table 1) was realized in order to compare experimental and theoretical results with varying crack position and size.

Due to the potential non-linear behaviour of the system it is important to use a white noise process to excite the beam in order to linearize the system response. Measuring the acceleration of the beam at several points (11 in this case) and acquiring the force acting at its free end, it is possible to evaluate all the Frequency Response Functions (Inertances) between these positions which lead to the extraction of the bending natural frequencies. The complete scheme of the measuring set-up is shown in Figure 1. All tests have been carried out using the LMS software Cada-X, applying the Least Squares Complex Exponential method in order to extract all the system poles and the Least Squares Frequency Domain procedure for estimating the system made shapes. The experimental natural frequencies of the nonslotted beams are listed in Table 2, while the experimental results obtained after the realisation of the slots are listed in Table 3.

4 Theoretical Results

For the studied test cases, a comparison of frequency shifts with the FEM and continuous model is presented in Tables 4 and 5 respectively. Figures 2 to 4 give a graphical representation of the normalised frequency variation maintaining the dimension of one of the cracks constant. Analysis of these curves show that, in general, there is good agreement between theoretical predictions and experimental results. The frequency shift becomes more important as the second crack depth is increased or if the cracks are closer to the built in end. Also the modes present different sensitivities as regards the slot existence depending on its location. The FEM model show improved results compared to the sensitivity model due to a more localized effect of the crack.

5 Conclusions

A test programme was carried out in order to obtain experimental results for a clamped-free beam with two cracks, to verify that accurate results can be simulated using two slots for experimental results and to give some insights regarding the correlation with FEM and sensitivity analytical models. The present work shows good agreement between the experimental results and the theoretical models for the studied cases. An acceptable description of the variation of the dynamic behaviour of a beam with two cracks can be obtained using slots as means of simulating cracks for double cracks. While the sensitivity models give a more global view upon dynamic behaviour, the FEM results are more accurate as regards frequency variation.

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beam	1st crack		2nd crack	
	pos	size	pos	size
1	120	2	200	2
2	120	2	200	4
3	120	2	200	6
4	120	2	200	8
5	120	2	480	2
6	120	2	480	4
7	120	2	480	6
8	120	2	480	8
9	120	4	480	2
10	120	6	480	2
11	120	8	480	2
12	400	2	480	2
13	400	4	480	2
14	400	6	460	2
15	400	8	480	2

Table 1: Experimental cases

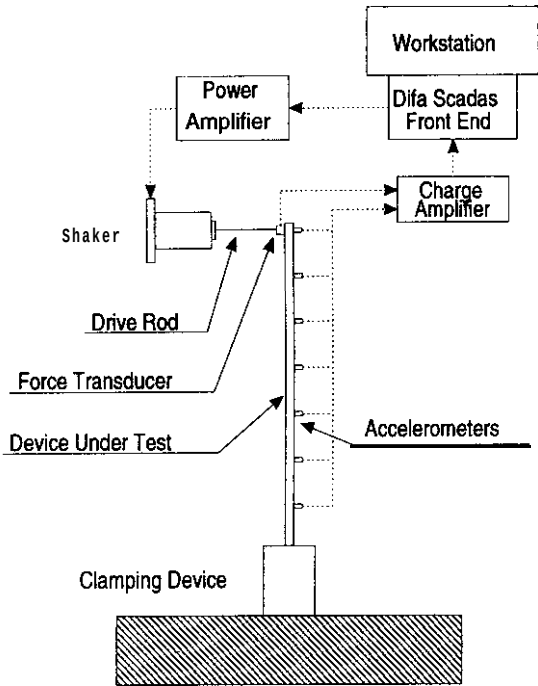


Figure 1: Test set-up

beam	f_1	f_2	f_3	f_4
1	24.25	153.25	425.25	845
2	24.25	153.25	425.25	845.75
3	24.25	153.25	425.5	845.25
4	24	153	423.5	843.75
5	24.25	152.75	424	843.5
6	24.25	153	424.25	844.75
7	24.25	153	424	845.25
8	24.25	153	424.5	845.75
9	24.25	153	424.25	844.75
10	24.25	153	424	845.25
11	24.25	153	424.5	845.75
12	24.25	153.25	425.25	845
13	24.25	153.25	425.25	845.75
14	24.25	153.25	425.5	845.75
15	24	153	423.5	843.75

Table 2: Experimental natural frequencies of the uncracked beams

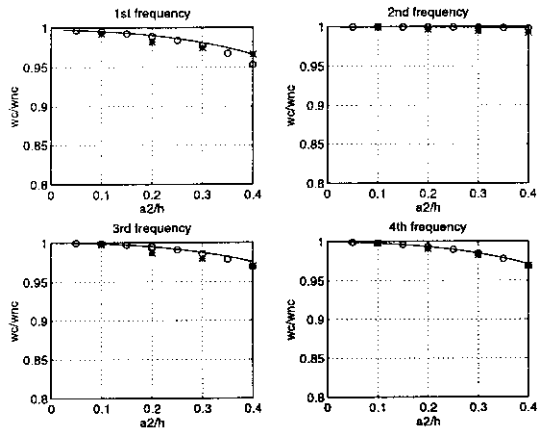


Figure 2: Frequencies comparison; $p_1=0.12\text{m}$, $p_2=0.20\text{m}$, $a_1/h=0.1$ (—) continuous model (o) FEM model (*) experimental data

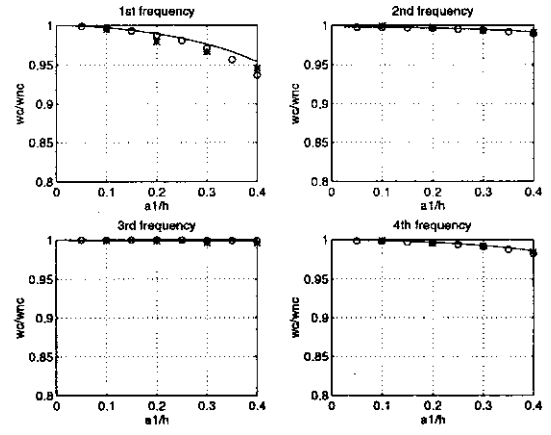


Figure 4: Frequencies comparison; $p_1=0.12\text{m}$, $p_2=0.48\text{m}$, $a_2/h=0.1$ (—) continuous model (o) FEM model (*) experimental data

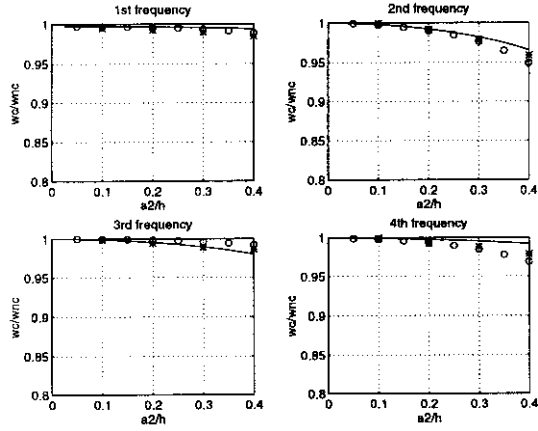


Figure 3: Frequencies comparison; $p_1=0.12\text{m}$, $p_2=0.48\text{m}$, $a_1/h=0.1$ (—) continuous model (o) FEM model (*) experimental data

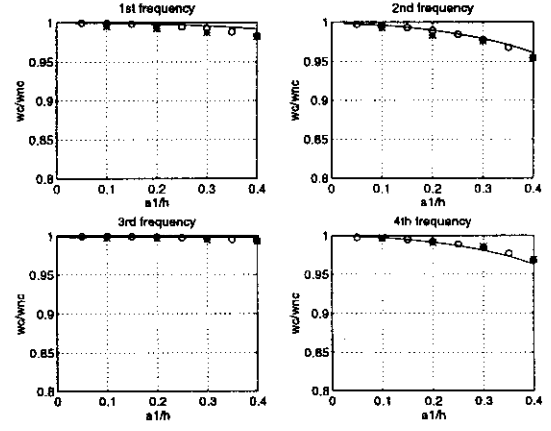


Figure 5: Frequencies comparison; $p_1=0.40\text{m}$, $p_2=0.48\text{m}$, $a_2/h=0.1$ (—) continuous model (o) FEM model (*) experimental data

beam	f_1	f_2	f_3	f_4
1	24.063	153.188	424.125	842.15
2	23.813	152.688	419.5	837.75
3	23.625	152.375	416.75	830.25
4	23.188	151.815	410.75	817.5
5	24.125	152.625	423.375	842.15
6	24.063	151.688	421.75	838.25
7	24	149.15	419.125	835.5
8	23.815	146.688	418.625	827.75
9	23.75	152.625	423.375	840.75
10	23.438	152.063	422.25	838
11	22.938	151.625	422.5	832.5
12	24.125	152.125	424.125	842.5
13	24.063	150.688	423.875	838
14	23.938	149.5	423.5	833.15
15	24.566	146.063	420.5	817.5

Table 3: Experimental natural frequencies of the cracked beams

beam	f_1	f_2	f_3	f_4
1	24.352	153.240	428.605	839.096
2	24.225	153.205	426.859	835.203
3	24.009	153.143	423.168	828.312
4	23.628	153.041	418.642	816.887
5	24.392	152.944	428.718	840.125
6	24.376	152.080	427.287	839.116
7	24.349	150.550	424.755	831.330
8	24.303	148.012	420.557	834.369
9	24.209	152.782	428.642	838.153
10	23.880	152.495	428.507	834.663
11	23.337	152.018	428.283	828.876
12	24.445	152.668	428.143	839.028
13	24.412	151.132	428.735	835.983
14	24.352	150.016	428.722	825.052
15	24.253	147.330	428.701	810.243
uncracked	24.463	153.309	429.253	841.187

Table 5: Theoretical natural frequencies of the cracked beams (continuous model)

beam	f_1	f_2	f_3	f_4
1	24.340	153.237	428.706	839.957
2	24.194	153.199	426.834	836.075
3	23.900	153.123	423.143	828.724
4	23.319	152.916	416.121	815.753
5	24.381	152.822	429.184	840.024
6	24.353	151.808	428.701	836.300
7	24.297	149.671	427.750	829.101
8	24.179	145.519	425.942	815.913
9	24.168	152.700	429.117	838.090
10	23.745	152.346	428.984	834.285
11	32.921	151.685	428.728	821.094
12	24.435	152.627	429.086	839.390
13	24.380	151.699	428.735	835.541
14	24.268	149.858	428.042	828.380
15	24.038	146.300	426.720	815.038
uncracked	24.463	153.239	428.725	840.013

Table 4: Theoretical natural frequencies of the cracked beams (FEM model)

uncracked beam			
0.9966	0.0219	0.0381	0.0381
0.0320	0.9880	0.0258	0.0218
0.0494	0.0235	0.9868	0.0251
0.0482	0.0566	0.0232	0.9795
beam no 11			
0.9963	0.0194	0.0163	0.0292
0.0289	0.9891	0.0282	0.0094
0.0547	0.0255	0.9851	0.0257
0.0390	0.0490	0.0234	0.9803
beam no 15			
0.9977	0.0163	0.0125	0.0187
0.0337	0.9915	0.0298	0.0163
0.0498	0.0233	0.9866	0.0353
0.0496	0.0443	0.0195	0.9841
beam no 8			
0.9988	0.0310	0.0199	0.0348
0.0298	0.9887	0.0220	0.0233
0.0364	0.0256	0.9883	0.0439
0.0308	0.0667	0.0325	0.9827

Table 6: Comparison by MAC of experimental and theoretical (FEM) mode shapes